Stirling's Approximation: Definition, Bounds, and Derivation

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1 Definition: Stirling's Approximation

Stirling's Approximation is an asymptotic formula used to estimate the value of large factorials. It states that as $n \to \infty$, the factorial function $n!$ grows at a rate proportional to $\sqrt{2\pi n}\left(\frac{n}{e}\right)^n$.

The General Form:

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$$

(The symbol $\sim$ means asymptotic equivalence: the ratio of the two sides approaches 1 as $n \to \infty$).

Taking the Natural Logarithm: In computer science and information theory, it is often more useful to work with the logarithm of the factorial:

$$\ln(n!) \sim n\ln(n) - n + \frac{1}{2}\ln(2\pi n)$$

Or, written using Big-O notation to show the dominant terms:

$$\ln(n!) = n\ln(n) - n + O(\ln n)$$

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2 Rigorous Bounds and Error Terms

Stirling's approximation is an *asymptotic* series. For small n, the approximation is not very accurate. However, the exact error can be bounded using the following rigorous inequalities for all $n \geq 1$:

$$\sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\frac{1}{12n+1}} < n! < \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{\frac{1}{12n}}$$

Robbins' Sharp Bound: A more symmetric and mathematically elegant bound was proven by Herbert Robbins in 1955:

$$\sqrt{2\pi}n^{n+\frac{1}{2}}e^{-n+\frac{1}{12n+1}} < n! < \sqrt{2\pi}n^{n+\frac{1}{2}}e^{-n+\frac{1}{12n}}$$

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3 The Higher-Order Series (Asymptotic Expansion)

Stirling's approximation is actually the first term of an asymptotic series. For higher precision, the full expansion can be used:

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n\left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{51840n^3} - O\left(\frac{1}{n^4}\right)\right)$$

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4 A Brief Derivation Sketch

The formula is derived by relating factorials to integrals (sums to areas).

1. The natural logarithm of a factorial is a sum: $\ln(n!) = \sum_{k=1}^{n} \ln(k)$.
2. This sum can be approximated by the integral $\int_1^n \ln(x)\,dx$.
3. Using integration by parts, $\int \ln(x)\,dx = x\ln(x) - x$. Evaluating from 1 to n gives $n\ln(n) - n + 1$.
4. This gives the leading terms: $\ln(n!) \approx n\ln(n) - n$.
5. When the error between the sum (rectangles) and the integral (area under the curve) is carefully analyzed using the Euler-Maclaurin formula, the first error term is $\frac{1}{2}\ln(2\pi n)$.
6. Exponentiating both sides yields the final formula.

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5 Example: $n = 10$

Test the approximation for $n = 10$.

- **Exact:** $10! = 3,628,800$
- **Stirling's Approximation:**

$$\sqrt{2\pi(10)}\left(\frac{10}{e}\right)^{10} \approx \sqrt{62.83} \times (3.6788)^{10} \approx 7.927 \times 458,361.3 \approx 3,633,860$$

- **Error:** The approximation is only about 0.14% off from the exact value. As n grows, this error shrinks to zero.

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6 Why is it used in Machine Learning and Combinatorics?

1. **Entropy and Information Theory:** In ML, the logarithm of binomial coefficients $\ln\binom{n}{k}$ often needs to be computed. Using Stirling's approximation, $\ln\binom{n}{k} \approx nH(k/n)$, where H is the binary entropy function. This connects probability to information theory.
2. **Discrepancy Theory:** It is used to estimate the probability of exactly hitting the mean (0 discrepancy) in a random walk.
3. **Statistical Mechanics:** The Boltzmann entropy formula $S = k\ln W$ relies on Stirling's approximation to transition from the discrete counting of microstates to continuous thermodynamics.